

Parallel Implementation of a 1D-CNN on GPU for Satellite Telemetry Anomaly Detection

Dario Liotta



UNIVERSITÀ
DEGLI STUDI
DI PADOVA



Dipartimento
di Fisica
e Astronomia
Galileo Galilei

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Anomaly Detection in Satellite Telemetry

Context

Satellites generate continuous multi-channel time-series telemetry to
monitor system health

Problem

Anomalies can occur, leading to critical system failures and loss of
spacecraft functionality

Idea

Employ a neural network model to automate time-series anomaly detection

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ESA Telemetry Benchmark

Our reference is real in-orbit operational time-series data provided by ESA

OPS-SAT CubeSat	{	303,493 temporal points
		2,123 continuous segments
		9 telemetry sensors
		1 s median sampling interval (Δt)
		Labels: $\sim 80\%$ nominal, $\sim 20\%$ anomalous

Originally designed for unsupervised anomaly detection in a reconstruction framework, the task is framed here as a **supervised binary classification** problem suited for a simpler model architecture

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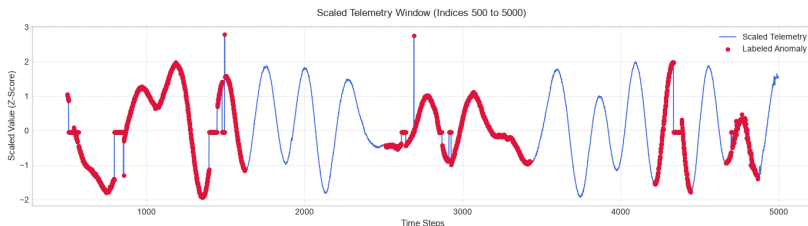
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Telemetry example

Slice of a continuous telemetry segment that shows **quasi-periodic oscillations**. Anomalies are highlighted in red, displaying strong deviations from the normal signal.



Preprocessing Pipeline

Key data preparation steps tailored for our classification framework:

- **Data splitting**

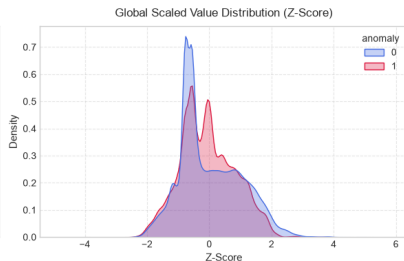
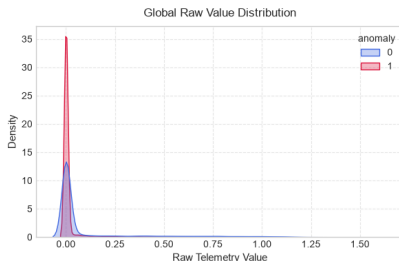
Split between training, validation and test sets at the segment level prior to further processing to prevent overlapping sequence leakage across sets.

Preprocessing Pipeline

Key data preparation steps tailored for our classification framework:

- **Channel-wise normalization**

Standardize each sensor independently using empirical mean μ and standard deviation σ estimated on nominal training data to avoid data leakage.

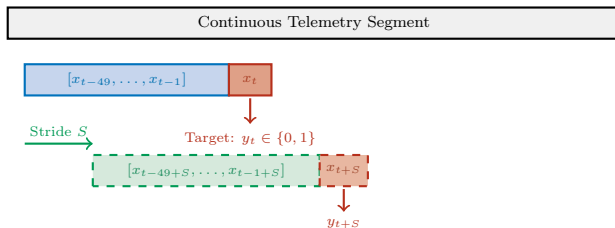


Preprocessing Pipeline

Key data preparation steps tailored for our classification framework:

- **Sliding window**

Extract fixed-length ($W = 50$) temporal windows within segments, mapping sequence $(x_{t-49}, \dots, x_t)$ to target label $y_t \in \{0, 1\}$ with stride $S = 5$ for training (to reduce redundancy) and $S = 1$ for validation and test



Thank you for your attention!